MO-ALA (MS 4584712) - Supplementary Tables S1-S5 (revision, 2026-10-21)

All numbers below are recomputed programmatically from the archived run archives (terrain_paths.mat, path_feasibility_audit_1016.mat, ablation_v8_degradation_1016.mat, dim_scaling_1016.mat); none is transcribed by hand.

Table S1. Hard-constraint audit of the published smoothed trajectories

Re-validated against the scenario terrains regenerated with the exact seeds used for the reported experiments (rng(2026+100) mountain, rng(2026+200) debris flow; H_{min} = 40, H_{max} = 120). Thresholds: maximum climb angle $\alpha_{max} = \pi/3$, minimum clearance h_{min} = 2 m, flight box [0,100] × [0,100]. Trajectories are split into the B-spline-smoothed ones (>= 50 points) and the discrete-waypoint fallback (<= 30 points, i.e. the optimiser's own output).

Scenario	Algorithm	Trajectories	Smoothed / fallback	Line segments	Overall pitch rate (%)	Pitch rate, smoothed (%)	Pitch rate, fallback (%)	Max over-limit (deg)	Mean over-limit (deg)	Ground-collision points
Mountain (S1)	MOALA	1500	1001 / 499	109578	1.459	1.614	0.000	27.31	13.36	0
Mountain (S1)	NSGAII	900	272 / 628	40116	1.030	1.534	0.000	26.26	9.54	0
Mountain (S1)	MOPSO	2814	158 / 2656	71418	0.367	1.675	0.000	27.03	7.96	0
Mountain (S1)	EALA	570	44 / 526	15402	0.643	2.273	0.000	24.74	11.75	0
Mountain (S1)	IALA	1941	1602 / 339	165717	1.749	1.828	0.000	25.75	13.13	0
Mountain (S1)	HALA	740	209 / 531	31842	2.029	3.122	0.000	27.23	13.13	0
Debris flow (S2)	MOALA	1500	943 / 557	105054	0.722	0.810	0.017	22.29	13.09	0
Debris flow (S2)	NSGAII	899	279 / 620	40641	0.327	0.482	0.000	20.72	9.23	0
Debris flow (S2)	MOPSO	2813	173 / 2640	72567	0.081	0.344	0.000	17.02	5.75	0
Debris flow (S2)	EALA	570	144 / 426	23202	0.220	0.358	0.000	7.08	3.62	0
Debris flow (S2)	IALA	2061	1867 / 194	188907	0.577	0.590	0.000	22.51	10.51	0
Debris flow (S2)	HALA	747	315 / 432	40257	0.817	1.055	0.000	22.19	13.23	0

Reading: Three pitch-rate definitions are reported, because they answer different questions: **Overall** = offending segments / *all* published segments (smoothed and fallback together; this is the quantity quoted in Section 2.1.5 and in the responses to Reviewers 1, 2 and 3); **smoothed** = restricted to the smoothed subset; **fallback** = restricted to the optimiser's own discrete waypoints. (i) ground-collision and out-of-bound violations are **exactly zero** for every algorithm, scenario and run (the out-of-bound column is omitted from the table below because it is identically zero in every run; the per-run values are in data/path_feasibility_audit_1016.mat); (ii) the **optimiser's own (unsmoothed) trajectories satisfy the pitch constraint at every segment but one** - the discrete-fallback pitch rate is 0.000 % for every algorithm and scenario (the only exception being a single segment out of ~6 000 for MO-ALA in the debris-flow scenario, 0.017 %); (iii) every residual over-limit is therefore introduced by the B-spline *display* post-processing, it is of the same order for every algorithm (MO-ALA 1.46 % / 0.72 % overall, 1.61 % / 0.81 % on the smoothed subset alone, is below HALA 2.03 % / 0.82 % overall and 3.12 % / 1.06 % smoothed, and comparable to NSGA-II 1.03 % / 0.33 % overall), and it is concentrated in the initial climb-out (about two thirds of the offending segments lie in the first decile of a trajectory). The residual pitch over-limits are therefore neither an MO-ALA anomaly nor evidence that the optimiser violates the constraint.

Table S2. SBX-hybrid operator vs. the original memory-guided operator

Identical protocol for both rows of each benchmark: $N = 30$, $T = 500$, $N_{arch} = 50$, 30 runs, paired by seed ($20260812 + 4000 \cdot b + 100 \cdot a + r$); two-sided paired Wilcoxon signed-rank.

Benchmark	HV original	HV hybrid	dHV (*p*)	IGD original	IGD hybrid	dIGD (*p*)
ZDT1	0.6541	0.6538	-0.0% (0.837)	0.0093	0.0100	+8.3% (6.98e-06)
ZDT2	0.3224	0.3201	-0.7% (0.027)	0.0091	0.0101	+11.4% (1.73e-06)
ZDT3	0.7788	0.7782	-0.1% (0.959)	0.0104	0.0214	+105.0% (5.75e-06)
DTLZ1	0.0000	0.0059	n/a (0.00195)	40.4788	1.0776	-97.3% (1.73e-06)
DTLZ2	0.4893	0.4899	+0.1% (0.651)	0.0951	0.0912	-4.0% (0.000771)
DTLZ3	0.0000	0.0420	n/a (0.00391)	108.8292	1.6796	-98.5% (1.73e-06)

Reading: on the two benchmarks with a deceptive/far-from-Pareto start (DTLZ1, DTLZ3) the SBX hybrid lifts the archive from an empty/near-empty front (HV 0.0000) to a usable one (0.0059 / 0.0420) and cuts IGD by ~97-99 %. On the smooth ZDT fronts it is weakly *worse* (IGD +8.3 % / +11.4 % / +105.0 %). This is the empirical basis for keeping the SBX hybrid as an auxiliary exploratory option rather than integrating it into MO-ALA.

Note on the dHV column: for DTLZ1 and DTLZ3 the relative HV change is reported as n/a because the original memory-guided operator produced an empty archive (HV =

0.0000), so a relative change is not defined. The paired Wilcoxon signed-rank p-value in parentheses is computed on the 30 per-run HV values and is therefore unaffected.

Table S3. Sensitivity of the TOPSIS selection to the risk weight w_{J_3}(MO-ALA)

The base late-stage weight vector $\boldsymbol{w}$ = (0.1, 0.1, 0.6, 0.2) has its risk weight w_{J_3} swept over {0.2 … 0.7}; the remaining weights are renormalised to sum to 1-w_{J_3}. The selection uses TOPSIS with per-front min-max normalization, the same procedure as Section 3.6; the w_{J_3} = 0.60 row therefore reproduces the late-stage numbers of Section 3.6. Means over the 30 runs.

w_{J_3}	Mountain: J_1 (/m)	Mountain: J_3	Mountain: C_i	Debris flow: J_1 (/m)	Debris flow: J_3	Debris flow: C_i
0.20	218.30	577.94	0.7319	239.61	566.19	0.7306
0.30	246.20	279.53	0.6940	250.48	267.95	0.6954
0.40	255.98	210.23	0.7359	253.78	186.52	0.7401
0.50	259.38	195.24	0.7866	257.57	169.89	0.7902
0.60	265.92	178.55	0.8372	260.04	157.96	0.8384
0.70	268.82	173.70	0.8815	263.67	147.38	0.8821

Reading: raising w_{J_3}. shifts the selected solution monotonically from short/high-risk to long/low-risk while the closeness coefficient remains high across the whole sweep (0.69-0.88), i.e. the mission-stage interface is monotone and smooth rather than a knife-edge.

Table S4. Consistently selected solution per algorithm (TOPSIS, late-stage weights $\boldsymbol{w}$ = (0.1, 0.1, 0.6, 0.2)); mean +/- std over the 30 runs

The selection uses TOPSIS with per-front min-max normalization, the same procedure as Section 3.6.

Scenario	Algorithm	J_1 (/m)	J_2	J_3	J_4 (/rad)	C_i
Mountain (S1)	MOALA	265.92 +/- 65.12	2.141 +/- 0.721	178.55 +/- 83.61	2.797 +/- 0.732	0.8372 +/- 0.0482
Mountain (S1)	NSGAII	429.63 +/- 80.44	2.998 +/- 0.711	291.02 +/- 47.89	2.968 +/- 0.588	0.7480 +/- 0.0245
Mountain (S1)	MOPSO	622.41 +/- 72.79	4.914 +/- 0.557	331.42 +/- 39.06	4.638 +/- 0.341	0.7501 +/- 0.0215
Mountain (S1)	EALA	428.36 +/- 210.79	3.507 +/- 1.802	753.35 +/- 105.83	2.985 +/- 1.591	0.7263 +/- 0.0416
Mountain (S1)	IALA	275.03 +/- 37.85	2.045 +/- 0.298	161.99 +/- 34.11	2.680 +/- 0.386	0.8372 +/- 0.0465
Mountain (S1)	HALA	201.85 +/- 59.68	1.357 +/- 0.591	90.07 +/- 65.61	2.201 +/- 1.057	0.7804 +/- 0.0570

Scenario	Algorithm	J_1 (/m)	J_2	J_3	J_4 (/rad)	C_i
Debris flow (S2)	MOALA	260.04 +/- 86.25	2.020 +/- 0.793	157.96 +/- 77.87	2.702 +/- 0.774	0.8384 +/- 0.0514
Debris flow (S2)	NSGAII	401.62 +/- 63.18	2.838 +/- 0.447	243.53 +/- 46.35	3.202 +/- 0.472	0.7405 +/- 0.0244
Debris flow (S2)	MOPSO	587.89 +/- 62.23	4.673 +/- 0.486	317.26 +/- 33.05	4.494 +/- 0.375	0.7600 +/- 0.0240
Debris flow (S2)	EALA	528.30 +/- 246.32	4.351 +/- 2.145	789.88 +/- 87.48	3.438 +/- 1.489	0.7396 +/- 0.0558
Debris flow (S2)	IALA	230.21 +/- 42.39	1.766 +/- 0.324	109.97 +/- 26.03	2.661 +/- 0.456	0.8537 +/- 0.0609
Debris flow (S2)	HALA	182.47 +/- 54.74	1.330 +/- 0.588	72.91 +/- 36.92	2.323 +/- 1.153	0.7877 +/- 0.0647

Note on EALA: the archived front is numerically degenerate (all front members identical) in 4 of the 30 mountain runs and 8 of the 30 debris-flow runs; C_i is undefined in those runs, so the C_i entries are means over the valid runs only (26/30 and 22/30, respectively), while the objective columns use all 30 runs.

Reading: each row is one consistently selected trajectory per run, so the length-versus-safety trade-off can be read within a row - unlike per-objective minima, which may belong to different members of the front.

Table S5. Wall-clock time vs. problem dimension (MO-ALA, mountain scenario)

$N = 30$, $T = 500$, $N_{arch} = 50$, one run per dimension.

N_w (waypoints)	$D = 3\ N_w$	Wall-clock (s)	Per-iteration (ms)	HV
10	30	33.79	67.576	0.1386
20	60	52.97	105.942	0.1106
30	90	66.31	132.619	0.0662

Reading: tripling the decision dimension (30 -> 90) raises the wall-clock only 1.96x (33.79 s -> 66.31 s), and the per-iteration increment decelerates (38.4 ms, then 26.7 ms), i.e. the cost grows sub-linearly in D at a fixed generation budget. This is consistent with an $O(N{\cdot}D)$ term superimposed on a D-independent fixed cost, and inconsistent with a quadratic dependence on the number of waypoints, which would have accelerated rather than decelerated.

Note on the HV column: each row's hypervolume is computed against a reference point estimated from that row's own front (MetricsCalculator.estimateHVRefPoint) and the front is min-max normalised to the unit box before the indicator is evaluated, exactly as for the other HV values in this paper. The three HV entries therefore share no common reference box and are not comparable across rows; they are listed only to document what each run produced, and no conclusion about the effect of dimension on solution quality is drawn from them. The wall-clock and per-iteration columns are computed on the same machine in the same session and are directly comparable.